Expt No: 09

Implementation of Subnetting in CISCO packet Tracer Simulator

Aim:

To implement the subnetting in Cisco Packet Tracer simulator.

Procedure:

i) Open Cisco Packet Tracer and create a new network topology

ii) Add the required routers, switches and PCs to workspace

iii) Connect the device using appropriate copper cables

iv) Use the network 192.168.1.0/24 and divide into /27 network

v) Configure the router interface with the required IP address and subnet mask 255.255.255.224

vi) Use enable and config terminal in the router CLI

vii) Enter the interface configuration terminal and assign IP using
   i) ip address <IP address> 255.255.255.224
   ii) no shutdown

viii) Configure the required switch port in access mode.

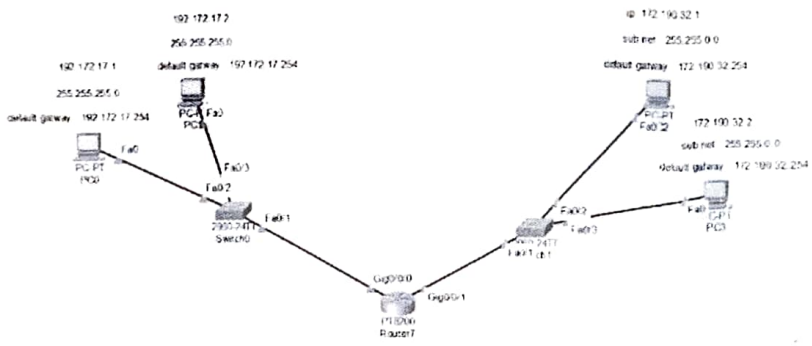
ix) Assign each PC its IP address, subnet mask and default gateway.

x) Repeat the configuration all required subnet

xi) Use the ping command to test connectivity between PC's and the router.

The ping is successful, the network is configured successfully.

```
Router>
*Sep 24, 11:17:27.1717: %LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0/1, changed state to up
Router>
Router>
Router>
Router>
Router>
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.172.17.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)# ip address 172.190.32.254 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
```



Result:

Thus subnetting is implemented using Cisco packet Tracer.